The DEGRACE ECOSMOG suite

What is on disk, what is pinned, and what MG-GLAM companions need

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Executive summary

Pinned from the files on disk

- 96 nodes of a Sobol design in $(\Omega_m, h, S_8, n_s, \log_{10}|f_{R0}|)$; every derived parameter reproduced exactly.
- f(R) (cosma7) and ΛCDM (cosma8) twins from the *same* 2LPTic ICs; seed 2026 for all nodes.
- 43 outputs per run: 17 primary z + 25 stations at $\chi = 256\,k\,h^{-1}\mathrm{Mpc}$ in each node's own background, verified to 2×10^{-7}.
- All 192 runs reach $a = 1$; Gadget snapshots and f(R) $P(k)$ at every epoch.

Two things to raise with Baojiu

- CAMB was fed $\omega_c = \Omega_m h^2$: linear spectra have $\Omega_m^{\mathrm{CAMB}} = \Omega_m + \Omega_b + \Omega_\nu$, 0.04–0.055 too high.
- Nodes 1–80 carry two byte-different Gadget conversions (15 and 27 March).

96 nodes, 1024^3 in $1024\,h^{-1}\mathrm{Mpc}$, $z_{\mathrm{init}} = 49$

192 complete runs (f(R) + ΛCDM twins)

43 outputs per run, per-node epoch lists

0 consistency failures over 96 nodes

Reference: `docs/degrace_ecosmog/` (README, `nodes.csv`, `aout.json`, extraction script).

Source: bi267 directories on cosma7 (dp004) and cosma8 (dp203), read 2026-10-16

Where everything lives

C7 = /cosma7/data/dp004/bl267/Runs/DEGRACE C8 = /cosma8/data/dp203/bl267/Projects/Ongoing/DEGRACE

item	path	notes
design, generators, CAMB	C7/ICs/CAMB/Parameter_files/Python_codes/	DEGRACE_L512.csv, Run_CAMB.py, Run_2LPTic.py, Gen_ECOSMOG_Params.py, params_nXXX.ini
initial conditions	C8/ICs/IC_data/L1024/Node_XXX/ics.0..27	30 GB per node; CAMB $P(k)$ and 2LPTic logs beside them
ΛCDM runs	C8/Runs/LCDM/Node_XXX/	namelist, info files, gadget_data
f(R) runs	C7/Runs/L1024/FR/Node_XXX/	same layout, restarts
f(R) $P(k)$	C7/Runs/L1024/FR/power_spectra/Node_XXX/	POWMES, 43 epochs
fiducial node	C7/Runs/L1024/FR/Node_900/	Planck (Ω_m 0.3089, h 0.6774); job script only, no ICs
tests	C8/Runs/L512/Node_002, C8/Runs/L256	node-2 cosmology, ΛCDM + f(R)
not run	C7/Runs/L1024/GR/Np2048/Node_002	ICs L1024h never generated
obsolete	C8/ICs/IC_data/L1024_old/	earlier design (101 nodes): do not use

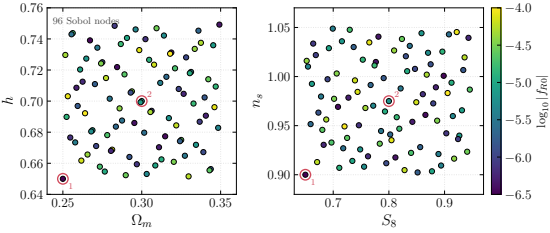
Nodes 097–100 are empty ΛCDM run directories (namelist only); the generators loop over 1–96. The suite is 96 nodes. Raw RAMSES particle files were deleted after conversion: only info_000NN.txt remains per output.

The design: a Sobol sequence in five parameters

col	parameter	range (nodes 1–96)
1	Ω_m	0.25 – 0.349
2	h	0.65 – 0.749
3	S_8	0.65 – 0.945
4	n_s	0.90 – 1.049
5	$\log_{10}\lvert f_{R0}\rvert$	-6.5 – -4.02
6	unused	-0.45 – 0.98
7–12	unit-cube coordinates	

Derived, verified for all 96 nodes

$\Omega_b = 0.02237/h^2$, flat, $\Omega_\Lambda = 1 - \Omega_m$
$\sigma_8 = S_8\sqrt{0.3/\Omega_m}$ (ICs within 10^{-4}: CAMB A_s tuning tolerance)
$f_{R0} = -10^{\text{col }5}$, Hu–Sawicki $n = 1$
ECOSMOG `fr_c1oc22` $= \lvert f_{R0}\rvert\,(12/\Omega_m - 9)^2$ (to 10^{-9})
`dens_V_on` = 80 / 120 / 160 for $\lvert f_{R0}\rvert < 10^{-6}$ / $< 10^{-5}$ / else



Node 1 is the lower corner of every range, node 2 the centre; colour is $\log_{10}\lvert f_{R0}\rvert$.

Source: `DEGRACE_L512.csv` (512 rows); `figs/make_suite_fig.py`

Initial conditions: 2LPTic, seed 2026 for every node

setting	value
box, particles	$1024\,h^{-1}\mathrm{Mpc}$, 1024^3 (`Nmesh` 1024)
glass	one-particle glass tiled $1024\times$: regular grid
start	$z = 49$, second-order LPT
seed	**2026 for every node**
spectrum	CAMB $z{=}0$ table as $(\log_{10} k,\ \log_{10} 4\pi k^3 P)$
normalisation	rescaled to `Sigma8` (`Normfac` in the log)
tilt	`PrimordialIndex = 1` (none beyond CAMB's n_s)
units	Mpc/h, $10^{10}\,M_\odot/h$, km/s
files	Gadget-1, little-endian, 28 files, 30 GB per node
growth factor	`inputspec_ics.txt` line 1: $(49,\ D_+$ ratio$)$

Why this matters for companions

- f(R) and ΛCDM of a node share the same IC files: matched-phase twins.
- Fixed seed across the design: cosmology dependence is not confused with sample variance.
- MG-GLAM can ingest the identical files with `ic2pm` (default `half` velocity epoch), giving a third matched run.

Checks on disk

RAMSES `info` headers (H_0, Ω_m, Ω_Λ) agree with `2LPTic.param` for all nodes; Gadget header: Box 1024, $a = 0.02$, 1024^3, mass 6.94 (node 1). POWMES $P(k)$ of every IC set at $z = 49$ (`PK_LCDM_NXXX_S01.dat`).

Source: `2LPTic.param`, `log_2LPTic.txt`, `Run_2LPTic.py`

Linear spectra: CAMB settings and the ω_c pitfall

Template (Planck 2018 base), per-node overrides

$\omega_b = 0.02237$, $\omega_\nu = 0.0006442$ (one 0.06 eV neutrino, $N_{\rm eff} = 3.046$), $\Omega_k = 0$, $w = -1$, $\tau = 0.0544$, $T_{\rm CMB} = 2.7255$, $Y_{\rm He} = 0.24$, pivot 0.05 Mpc^{-1}, no running, $k_{\max} = 20$, 5 per log-int, linear, $z = 0$ only.

Per node: `hubble`, `omch2`, n_s; A_s iterated until $\sigma_8(z{=}0)$ hits $S_8\sqrt{0.3/\Omega_m}$ within 10^{-4}.

Files

`PK_CAMB_z00_XXX.dat`: k/h, $P(k)$ $[(\mathrm{Mpc}/h)^3]$, 617 rows, $z = 0$. The copy beside the ICs is byte-identical to the cosma7 original.

Pitfall in `Run_CAMB.py`

The script computes `Oc = Om - Ob` but writes `omch2 = Om*h0**2`. CAMB's total matter is therefore

$$\Omega_m^{\rm CAMB} = \Omega_m + \Omega_b + \Omega_\nu = \Omega_m + (0.041 \ldots 0.055),$$

while 2LPTic, ECOSMOG and the D_+ scaling to $z = 49$ use the design Ω_m. The *shape* of every linear spectrum belongs to a higher ω_m; the *amplitude* is still the design σ_8 because 2LPTic renormalises.

Harmless for companions that ingest the same ICs. Not harmless for emulator labels, "linear" ratios or halo-model inputs built from these files. `nodes.csv` carries both Ω_m values.

Source: `Run_CAMB.py` lines 108–112; `params_n001.ini`: omch2 = 0.10563 = 0.25×0.65^2

ECOSMOG configuration and run status

namelist item	value
`levelmin/levelmax`	10 / 18: 1024^3 root grid, 8 refinements, 3.9 kpc/h finest cell
`m_refine`	4,4,4,8,...; `nsubcycle` 1,2,2,...; `nremap` 10
`ngridtot/nparttot`	3×10^9 / 1.5×10^9
cosmology	from the Gadget IC header
ΛCDM twin	same namelist, `extradof=.false.`
job	560 MPI ranks, full nodes, 72 h, restarts

Status

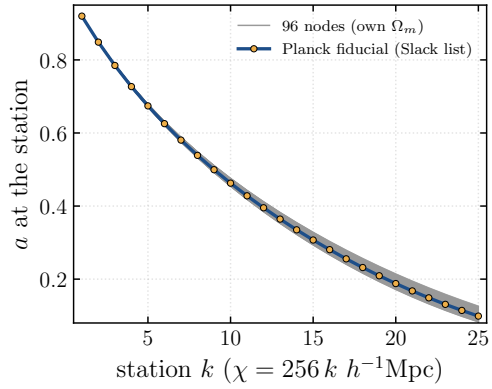
96/96 ΛCDM (cosma8, dp203) and 96/96 f(R) (cosma7, dp004) have all 43 outputs and end at $a = 1$.

Extras found

- **Node_900**, the Planck fiducial (Ω_m 0.3089, h 0.6774, Ω_b 0.0486): directory and job script created 8 Oct 2026, no ICs, not run. Its output list is the Slack list.
- **L512 node 2**: ΛCDM and f(R) test runs, 43 outputs each; **L256** ΛCDM test.
- **GR/Np2048 node 2**: 2048^3 ΛCDM namelist prepared, ICs never generated.
- **L1024_old**: 101 IC sets of an earlier design (node 1: Ω_m 0.333, h 0.731). Superseded.

Source: `namelist/cosmo.nml`, `run_cosma8.bash`, `info_000NN.txt`

Output epochs: the Slack convention, per node



Generator (`Gen_ECOSMOG_Params.py`)

43 outputs = `output_00001` at $a = 0.02$ plus 42 `aout`: $z = 0.125 \ldots 2.0$, $a = 1$, and the 25 roots of $\chi(a) = 256\,k\,h^{-1}\mathrm{Mpc}$, with $c/H_0 = 2997.92458\,h^{-1}\mathrm{Mpc}$ and $E^2 = \Omega_m a^{-3} + 1 - \Omega_m$ (the node's own Ω_m; matter and Λ only).

Verified

Recomputing the stations reproduces all 96×25 values to 2.3×10^{-7} in a; 17 primary + 25 secondary in every list; f(R) and ΛCDM namelists carry identical lists.

Every node has 1–4 neighbouring targets closer than 0.5 % in a: the exact-z scheduler accepts them since `cz ae4fa9d`. ECOSMOG itself dumps at the first step past each target, so actual snapshot epochs carry small offsets.

Snapshot products

Gadget conversions (r2g)

`gadget_data/output_000NN/gadget.0` …`gadget.559`, one file per rank, with `r2g.config.snap_NN` and a log. Little-endian Gadget-1, positions in **kpc/h** (BoxSize 1023999.98), masses $6.94 \times 10^{10}\, M_\odot/h$, IDs $1 \ldots 1024^3$. 26–29 GB per snapshot, about 1.2 TB per node and gravity model.

Two conversions

Nodes 001–080 also have `gadget_data_old` (15 March; the 27 March files are byte-different). Nodes 081–096 were converted once, later. Use `gadget_data`; ask what changed in r2g.

Power spectra

f(R): POWMES on every snapshot, `PK_FR_NXXX_SNN.dat`, 1024^3 grid with folding ($k = \mathrm{col}_1 \times 2\pi/L$, $P = \mathrm{col}_4 \times L^3$ as in Baojiu's `plot.py`). ΛCDM: only the IC-level $P(k)$ at $z = 49$; no snapshot spectra found on cosma8.

Usable for comparisons

- f(R)/ΛCDM snapshot pairs at 43 common epochs: halo catalogues, lens planes, $P(k)$ ratios.
- f(R) $P(k)$ ready-made at every epoch; ΛCDM $P(k)$ must be measured by us.

Source: Gadget headers read 2026-10-16; `du` of `output_00043` (26 G cosma8, 29 G cosma7)

Recipe for an MG-GLAM companion of node X

MG-GLAM input	source
initial conditions	`ICs/IC_data/L1024/Node_X/ics.*` through `ic2pm` (default `half`); Box 1024, 1024^3, $z_{\rm init} = 49$
cosmology	`nodes.csv`: Ω_m, h, $\Omega_b = 0.02237/h^2$, σ_8 = `sigma8_ic`, n_s
linear table	`PK_CAMB_z00_X.dat` normalised to `sigma8_ic`, with the $\Omega_m^{\rm CAMB}$ caveat
f(R)	Hu–Sawicki $n = 1$, `fR0` from `nodes.csv` (negative); twin with `MG_flag = 0`
outputs	`#outputs = -42`, $z = 1/a - 1$ from `aout.json` (exact-z mode)
resolution	ECOSMOG resolves 3.9 kpc/h. GLAM: $N_g = 2048$ (0.5 Mpc/h) or 4096 (0.25 Mpc/h) on the same 1024^3 particles. A 2048^3 run needs new ICs; whether 2LPTic's seed table keeps the large-scale modes at higher `Nmesh` is unverified
comparison data	f(R) $P(k)$ at 43 epochs; Gadget snapshots of both gravities

Already in place on `cz`

`ic2pm` with the half-step velocity epoch; exact-z outputs including close pairs (`ae4fa9d`); lens-plane plan (`docs/degrace_companion_implementation_plan.md`).

Still to build

Per-node `Init.dat`/`PkTable.dat` generator reading `nodes.csv` and `aout.json`; lens planes (I2) and the schedule index (I4); a pilot on one node against the ECOSMOG twin.

Open questions for Baojiu and next steps

Questions

1. Was `omch2` $= \Omega_m h^2$ in the CAMB inputs intended?
2. What changed between the 15 and 27 March r2g conversions; is `gadget_data_old` disposable?
3. r2g velocity convention: Gadget $\sqrt{a}$ factor or peculiar km/s?
4. Node_900: f(R) or ΛCDM, which f_{R0}, when?
5. Are Node_097–100 and `L1024_old` disposable?
6. Will ΛCDM snapshot $P(k)$ be measured, or do we measure both sides?

Next steps on our side

1. Send Baojiu the cross-check and the questions.
2. Write the per-node GLAM input generator from `nodes.csv` / `aout.json`.
3. Pilot node 2 (design centre): ingest its ICs, run ΛCDM and f(R) with the 42-output schedule, compare $P(k)$ with the ECOSMOG twins at the common epochs.
4. Decide the companion resolution (N_g, particle count) from the pilot.

Everything on these slides is reproducible:
`docs/degrace_ecosmog/extract_metadata.py`
re-reads the directories and reports any drift.